

Building brewing water with dissolved chalk

From German brewing and more

In order to build brewing water with temporary hardness (alkalinity and calcium hardness), chalk (CaCO_3) is needed. Chalk, however, is not very soluble in water and most of what is added remains suspended or settles to the bottom. This article describes how chalk can be dissolved if so desired.

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About chalk and the carbonate system

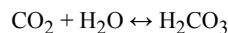
If you do not want to be bothered with the chemistry behind dissolving chalk you can safely skip ahead.

Chalk, also known as calcium carbonate (CaCO_3), is a salt of carbonic acid. Its ions are calcium (Ca^{2+}) and carbonate (CO_3^{2-}). If we know one thing about chalk in brewing it is that it doesn't dissolve very well in water. This is because the solubility product between calcium and carbonate is a very small number. Chemists write it like this:

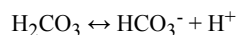
$$[\text{Ca}^{2+}] * [\text{CO}_3^{2-}] \leq K_{sp} = 3.36 \times 10^{-9}$$

The names in brackets ([]) stand for the concentration of the particular ions. What that formula means is that water can only hold calcium and carbonate in solution as long as the product of their concentrations is less or equal to 3.36×10^{-9} (0.0000000336). If the product is greater that calcium carbonate will precipitate until enough calcium and carbonate have been removed from the water to satisfy the equation.

But how can we get calcium carbonate dissolved in water? We do this by removing the carbonate that chalk adds when it dissolves. To understand this we have to take a look at carbonic acid and the carbonate system. Carbonic acid is formed when carbon dioxide is dissolved in water. Not all the dissolved carbon dioxide (CO_2) will form carbonic acid but some does according to this chemical equation:



The " $\leftrightarrow$ " means that there is a constant back and forth between both sides. I.e. carbonic acid is formed and falls apart into CO_2 and water constantly. But on average there is always a specific portion of CO_2 bound as carbonic acid. The interesting thing about carbonic acid is that it is a weak acid that can loose up to two protons (H^+). We looked at weak acids in Weak Acids and Bases where we saw that the loss or acceptance of protons depends on the pH of the environment. When a carbonic acid molecule looses one proton it becomes a bicarbonate ion:



This process is reversible. I.e. if a bicarbonate ion picks up a proton it becomes carbonic acid.

When a bicarbonate ion looses a proton it becomes carbonate:

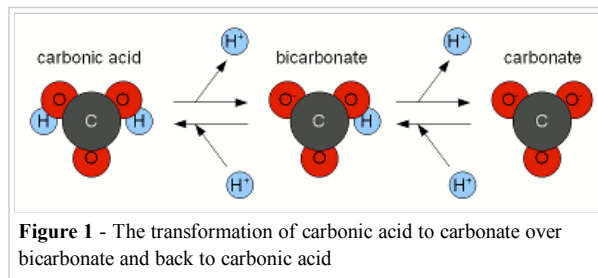
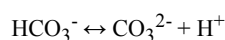


Figure 1 - The transformation of carbonic acid to carbonate over bicarbonate and back to carbonic acid

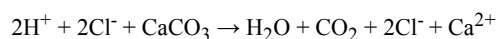


That process is illustrated in Figure 1. As mentioned earlier, the relative concentration of these 3 carbonate species, as they are called, [deLange] depends on pH since it is pH which determines how much H^+ is available. The more there is the more carbonic acid will exist and the less there is the more carbonate will be present. Figure 2 plots these relative concentration as % of the total carbonic acid + bicarbonate + carbonate amount over a pH range of 0 to 14.

But how does this tie into dissolving chalk?

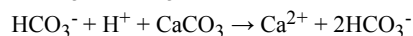
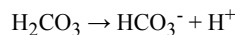
Well, earlier we saw that we need to remove the carbonate that dissolving chalk contributes to the water in order to allow the chalk to dissolve. This is done by converting the carbonate into bicarbonate. This process consumes protons. It is these protons that we have to supply to dissolve more chalk than the water can hold without any help.

To get these protons we need to add an acid. After all that's what an acid does, it adds protons and as a result lowers the pH of a solution. For acids we have basically two choices carbonic acid or other acids. Let's have a look at what happens when we add an acid other than carbonic acid. Hydrochloric acid for example which, in water, falls apart into a proton (H^+) and a chloride ion (Cl^-):



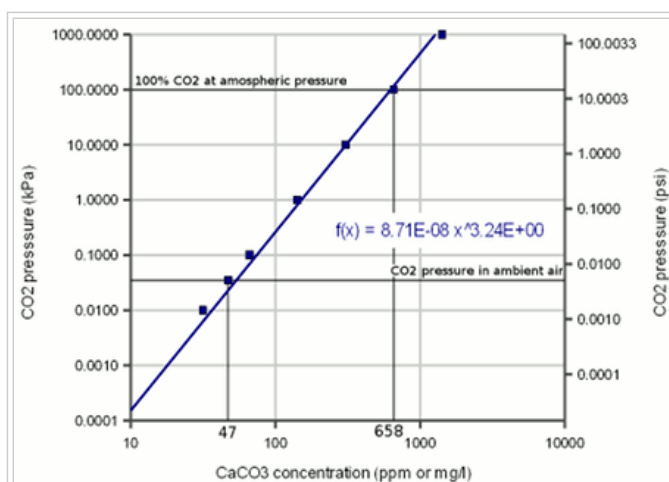
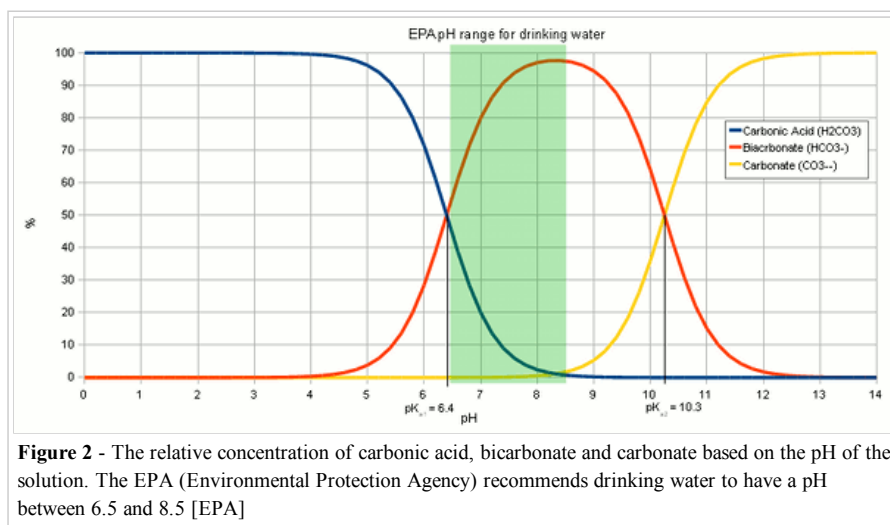
The result of the reaction between hydrochloric acid and the chalk is water, CO_2 (which escapes), chloride (Cl^-) and calcium (Ca^{2+}). Looking at these results, we could have done this water modification much simpler by just adding calcium chloride as a salt. One of the reasons why we want to dissolve chalk in water is increasing its alkalinity which is caused by bicarbonate and carbonate ions. If the bicarbonate or carbonate content of the water doesn't change its alkalinity won't change either.

Acids other than carbonic acid don't work. So what about carbonic acid. When carbonic acids becomes bicarbonate it loses a proton which can convert the carbonate ions added by the chalk to bicarbonate. Here is what this looks like:



The chalk molecule fell apart into a calcium ion and a bicarbonate ion. The other bicarbonate ion comes from the carbonic acids which lost a proton. Because this proton was lost to dissolving chalk we can say that the one molecule of chalk was also responsible for the creation of that bicarbonate ion. In the end one molecule of chalk, when dissolved with carbonic acid, creates one calcium ion and two bicarbonate ions.

Getting carbonic acid into water is as easy as adding CO_2 to it. We saw earlier how some of the CO_2 dissolved in water forms carbonic acid. To dissolve a substantial amount of chalk, however, much more CO_2 than the atmosphere is providing is needed. From carbonating beer we know that the CO_2 content of beer depends on its temperature and head space CO_2 pressure. The same is true for water: the higher the CO_2 pressure the higher the amount of dissolved CO_2 . Using a number of chemical equilibrium equations the relation between the CO_2 pressure over water can be correlated with the amount of chalk that can be dissolved in that water [deLange]. Figure 3 shows how the CO_2 pressure correlates with the amount of chalk water can hold. Note that both the x and the y axis are logarithmic and that the seemingly linear relationship is actually close to a cubic function. In practical terms that means that dissolving more than ~800 ppm $CaCO_3$ in water is unpractical since it requires a substantial amount of CO_2 pressure which we don't generally have access to. This needs to be kept in mind for the proposed procedure of dissolving chalk which initially dissolves all the chalk in a volume smaller than the actual brewing water volume.



Why dissolve chalk

Now that we know how to dissolve chalk in water we need to ask us why put forth this effort? Especially since the mash will provide an acid environment that should dissolve any chalk that wasn't dissolved in the water. At mash pH (~5.5) less than 0.00005 % of the carbo species are carbonate.

For some reason, however, mash pH experiments (http://braukaiser.com/documents/effect_of_water_and_grist_on_mash_pH.pdf) conducted with suspended and dissolved chalk showed that suspended, i.e. undissolved chalk, is limited in its ability to raise the mash pH. This is shown in Figure 4. Undissolved chalk was unable to raise the mash pH by more than 0.2 units. This was the case for both Pilsner and Munich malt mashes and does not seem to depend on the acidity of the malt. Dissolved chalk, on the other hand, showed a nice fairly linear relationship between chalk concentration and pH.

I tested this in a side-by-side experiment where I brewed a Schwarzbier (<http://braukaiser.com/lifestyle2/index.php?op=ViewArticle&articleId=132&blogId=1>) with water that had suspended chalk and water that had dissolved chalk. Based on the mash pH research I knew that, when dissolving it, I could use only half the chalk and would still get the same mash pH. That was validated in the experiment.

If you want to brew according to the German *Reinheitsgebot* you will have to dissolve the chalk, since the addition of salts in the mash or the use of water that doesn't meet drinking water standards is not allowed.

All this being said, dissolving chalk to build brewing water should be considered a technique for the advanced brewer. I have brewed many excellent beers by adding chalk directly to the mash or by not dissolving it in brewing water and only just recently started to dissolve the chalk for my brewing water. As with many other techniques, a brewer will have to see for him or herself if it is worth the additional effort.

How to dissolve chalk using carbonated water

To brew an upcoming batch of Kaiser Alt I want to mimic the water profile of the city of Düsseldorf where most of the Alts are brewed in Germany. One of the most well known Alt breweries, *Zum Uereige*, however, does not use city water. According to this on-line statement (<http://www.dooyoo.de/bier/uerige-alt/296389/>) the brewing water for that brewery comes from their own deep well. The Düsseldorf public water, on the other hand, is essentially cleaned and treated water from the river *Rhein*.

The current water report for Düsseldorf is available on-line: Water report Düsseldorf (<http://www.swd-ag.de/privatkunden/wasser/trinkwasseranalyse.php>) and upon calculating the residual alkalinity for that water one finds 108 ppm as CaCO_3 or 6.1 dH (dH is German Hardness). Such a residual alkalinity is well suited for brewing a dark amber beer like an Alt and I assume that most Alt breweries use city water without any water treatment.

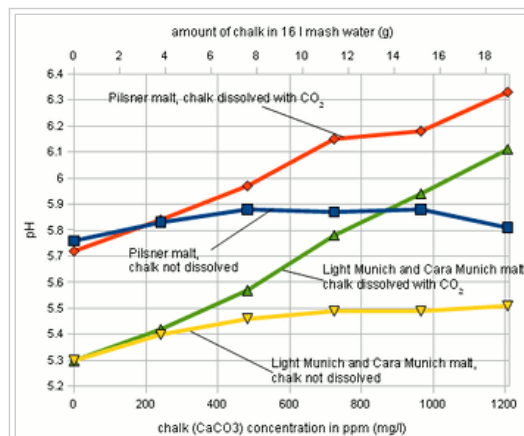


Figure 4 - The mash pH for mashes made with distilled water and various levels of chalk addition. One set of experiments dissolved the chalk with CO_2 while the other set left the chalk undissolved

The process step by step

Calculating the salt additions

In order to calculate the addition of dissolved chalk correctly you need a brewing water spreadsheet that supports dissolved chalk. Most of the available spreadsheets don't since they assume that each chalk molecule contributes only one bicarbonate ion. You can test this by adding 1 g of chalk to 1 l (3.78g to 1 gal) of water. If the resulting alkalinity (not residual alkalinity) is 490-500 ppm as CaCO_3 only half the chalk's alkalinity potential is considered and the chalk is not supposed to be dissolved in the water. My water spread sheet, *Kaiser_water_calculator.xls* (http://braukaiser.com/documents/Kaiser_water_calculator.xls), supports both dissolved and undissolved chalk in brewing water. In general, when dissolving chalk only half the amount of chalk is needed to build a water with the same residual alkalinity as a water build with undissolved chalk.

Step 1

On the "advanced" work sheet enter the starting water profile. I use reverse osmosis water which is very soft and for which I have a water analysis. If you are using reverse osmosis water and don't have an analysis simply leave the fields for the starting water profile 0. Due to the low ion concentration compared to the final ion concentration of the brewing water the error will be minimal and too small to matter.



base water and blending									
Water A percentage		100							
	Water A	Water B	analysis	mix					
calcium	5	0	mg/l	5.0	Ca mg/l	-0.6	ion balance (%)	0.93	hardness (dH)
magnesium	1	0	mg/l	1.0	Mg mg/l	0.7	Ca dH	0.2	Mg dH
sodium	5	0	mg/l	5.0	Na mg/l	0.6	Na dH	0.1	SO4 dH
sulfate	1	0	mg/l	1.0	SO4 mg/l	0.4	Cl dH		
chloride	5	0	mg/l	5.0	Cl mg/l				
bicarb	0	0	mg/l	24.4	HCO3 mg/l				
alkalinity	20	0	ppm CaCO3	20.0	CaCO3 mg/l	1.1	alkalinity dH	0.9	dH
Residual alkalinity				16	as CaCO3				

Step 2

With an eye on the resulting water profile (section below or right most column) enter concentrations of brewing salts. Rather than entering the final amounts of salts needed and considering the volume of water that will be treated, salts are entered as ppm (equivalent to mg/l). This makes the "water recipe" independent of the volume of water that needs to be treated. Once the water volume is known the spreadsheet calculates the required amounts of salts and displays them in a later section.

water modification using salts								
Gypsum	Epsom	Table Salt	Calcium Chloride	Magnesium Chloride	Baking soda	Chalk undissolved	Chalk dissolved	
CaSO4 · 2H2O	MgSO4 · 7H2O	NaCl	CaCl2 · 2H2O	MgCl2 · 6H2O	NaHCO3	CaCO3	CaCO3 + CO2	
ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm
20	110	0	135	0	120	0	85	
4.66			36.60			0.00	34.04	80
	10.84			0.00				12
		0.00			32.86			38
11.16	42.87							55
		0.00	65.10	0.00				70
					87.14	0.00	103.61	215
pH shift	0.15	RA (ppm CaCO3)			112.02		RA (dH)	5

Step 3

Check the resulting water profile. If you compare this profile to the Düsseldorf water report which was referenced earlier you'll find it to be a very close match. This is not always possible since we brewers don't have all combinations of cations (positive ions) and anions (negative ions), i.e. salts, available to us. In those cases focus on matching the residual alkalinity, calcium, sulfate and chloride content of the water.

resulting water profile									
range*									
50-150	80.5	Ca mg/l	14.01	hardness (dH)					
10-30	11.8	Mg mg/l	11.27	Ca dH	4.02	in Eq/l Ca			
0-150	37.9	Na mg/l	2.74	Mg dH	0.97	in Eq/l Mg			
0-350	55.0	SO4 mg/l	4.62	Na dH					
0-250	70.1	Cl mg/l	3.22	SO4 dH					
	215.1	HCO3 mg/l	5.55	Cl dH					
	176.4	as CaCO3							
			9.90	alkalinity dH	3.53	in Eq/l			
RA	112.02	as CaCO3							
pH shift	0.15		6.29	dH	2.24	in Eq/l			

Step 4

Now it is time to enter the total water volume and strike water volume. The spreadsheet will calculate the sparge water volume from these numbers. You should also enter the grist weight which will allow for the calculation of mash thickness. Mash thickness is important for making a mash pH estimation from beer color and residual alkalinity since mash thickness determines the pH shift that a given residual alkalinity causes.

Once you enter the beer color in SRM and how much of the specialty malts are roasted malts an estimate of the mash pH can be made. In this case it is 5.5 and well within the acceptable range of 5.2 - 5.7. The lower right hand corner of the mash pH section also shows estimates for the residual alkalinity that would be necessary to reach a mash pH of 5.2, 5.4 and 5.6 respectively. The large range from ~100 to ~200 ppm as CaCO₃ shows how wide of a range of brewing waters could be used to brew a 20 SRM beer with a mash thickness of 4 l/kg (~2 qt/lb)

Water use		
total water (liter)	29	l
Strike water (liter)	16	l
Sparge water (liter)	13	l

malt weight		
total grist weight	4	kg
mash thickness	4	l/kg

beer color based mash pH estimation						
beer color	20	SRM	target pH			
roasted %	10	%	5.2	5.4	5.6	
estimated mash pH	5.5	RA (as ppm CaCO3)	-99	55	209	

Step 5

Using the water volumes specified earlier, the amounts of salts needed are calculated. Because I don't have the ability to treat all the water at once I have to treat mash and sparge water separately. But for those able to treat all the needed water at once the amounts needed for the total volume are also given.

Of interest for dissolving chalk is the "dissolving chalk" section. Here you enter the volume of water that the chalk will be dissolved in and the spreadsheet estimates the minimal CO₂ pressure needed to dissolve the chalk. You'll quickly find that the relation between chalk concentration and pressure is not linear and that the pressure quickly shoots up once you exceed about 750-800 ppm. As a result this concentration of chalk is generally the practical limit and to dissolve more chalk you need to dissolve it in a larger amount of water.

If possible I like to dissolve my chalk additions in 2 l soda bottles and in this case it will be possible. Using 1.9 l to dissolve the mash water addition and 1.6 l to dissolve the sparge water addition I need only 1.55 bar (23 psi) and 1.38 bar (20 psi) respectively. Those pressure numbers are absolute which means that they include atmospheric pressure and that I have to use >0.55 bar (> 8 psi) and >0.38 bar (> 6 psi) as the setting on the regulator. A CO₂ pressure regulator measures pressure in excess of the atmospheric pressure.

necessary salt additions							
Gypsum	Epsom	Table Salt	Calcium Chloride	Magnesium Chloride	Baking soda	Chalk undissolved	Chalk dissolved
g	g	g	g	g	g	g	g
0.32	1.76	0.00	2.16	0.00	1.92	0.00	1.36
0.26	1.43	0.00	1.76	0.00	1.56	0.00	1.11
0.58	3.19	0.00	3.92	0.00	3.48	0.00	2.47

dissolving chalk				
	water	conc.	min CO2 pressure	
	l	ppm	bar	psi
mash	1.9	715.79	1.55	22.41
sparge	1.6	690.63	1.38	19.95
total		#DIV/0!	#DIV/0!	#DIV/0!

Building the brewing water



Figure 5 - These are the needed items: 2 liter soda bottles, carbonator cap, funnel, weighed salts and water. I mark the bottles with batch number and if the "brine" is for mash or sparge water treatment



Figure 6 - Set the funnel into the bottle and dump the salts into the funnel



Figure 7 - Add water. The water will wash any salts, that got stuck in the funnel, into the bottle. Even if close to 2 l water are needed, don't fill the bottle all the way yet. Leaving a headspace makes carbonating the water easier



Figure 8 - Add the carbonator cap, press it open and squeeze out all the air. Removing all the air and replacing it with CO₂ will make carbonation much more effective since the headspace will be completely filled with CO₂



Figure 9 - Hook up the CO₂ bottle via a regulator and shake the bottle to carbonate the water. Keep shaking until no more CO₂ is flowing into the bottle. Once carbonated top off with more water, if necessary, and carbonate again. Repeat the process for the other bottle(s)



Figure 10 - After standing for a while the water should clear. A slight haze may remain but there should not be a large amount of chalk sediment that turns the water cloudy once shaken up. If the water doesn't clear divide the contents evenly between two bottles, add water and carbonate again. Maybe the chalk concentration was too high to be dissolved by the available CO₂ pressure

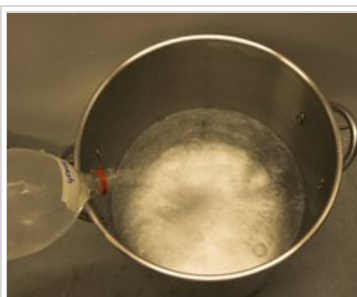


Figure 11 - The day before or on brewday I add the salt brine to the rest of the reverse osmosis water to get the desired volume of strike water and later the same for the sparge water. While CO₂ is now able to escape from the water the chalk will not come out of solution very quickly. It has been diluted so much, that it will take a long time for enough CO₂ to escape in order for chalk to precipitate again. You'll see more CO₂ escape while heating the water but since it is not boiled, enough CO₂ remains in solution to keep the chalk dissolved



Figure 12 - If you test the pH of the water you will find that it is lower than what you expect from fairly alkaline water. This low pH is the result of an excess of carbonic acid but the pH does not affect the alkalinity of the water. When I tested the hardness and alkalinity with a GH&KH test kit I found a hardness of 15 dH (270 ppm as CaCO₃) and an alkalinity of 10 dH (180 ppm as CaCO₃). Which matched the expectations close enough.

An alternative vessel for dissolving chalk are corny kegs. Their advantage is being able to hold larger volumes of water and as a result the chalk concentration can be much lower. With that the needed CO₂ pressure is substantially lower. The main disadvantage is the difficulty of visual inspection since it requires drawing a sample. But due the lower CO₂ pressure that is needed it is much more likely that all chalk gets dissolved in the larger water volume in a corny keg compared to the 2 liter limit in a soda bottle.

References

[deLange] A. J. deLange, Alkalinity, Hardness, Residual Alkalinity and Malt Phosphate: Factors in the Establishment of Mash pH (http://ajdel.wetnewf.org:81/Brewing_articles/Cerevesia/Final_galley), 2004
 [EPA] epa.gov, Secondary Drinking Water Regulations (<http://www.epa.gov/ogwdw000/consumer/2ndstandards.html>)

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